

Tyler Hickerson

📍 Santa Rosa, CA | ✉ tylerhickersonbe@gmail.com | ☎ 7077574685 | 💻 <https://tylerhbe.github.io/>

Education

California Polytechnic State University, San Luis Obispo (Cal Poly)

Bachelor of Science in Computer Science – Expected June 2028

GPA: ~3.67

Relevant Coursework: Data Structures, Object-Oriented Programming, Computer Systems, Discrete Structures, Systems Programming (C, UNIX), Algorithms, Assembly Language

Santa Rosa Junior College

~12 units completed, GPA: 4.0

Relevant Coursework: Programming in C++, Web Development (HTML, CSS, UX/UI principles)

Maria Carrillo High School, Santa Rosa, CA

Graduated June 2024 | GPA: 4.0 (Unweighted)

Relevant Coursework: AP Physics 1 & C (5 on exams), Honors Computer Science

Technical Skills

Programming Languages:

- **Proficient:** C, C++, Java, JavaScript, HTML/CSS
- **Experienced:** C#, Python, Bash, SQL, Assembly (MIPS/LC-3)

Frameworks & Libraries:

- MERN Stack (MongoDB, Express, React, Node.js)
- WPILib (FRC Robotics), PhotonVision (robotics vision), Steinberg VST3 SDK (audio plugins)
- Unity Game Engine (C#, 2D/3D development), Blender (3D modeling)

Systems & Tools:

- Unix/Linux environment, shell scripting, system calls, networking
- Redis (session caching), WebSockets (real-time data), Git/GitHub (version control)
- yt-dlp, ffmpeg, Raspberry Pi hardware integration

Core Competencies:

- Data Structures & Algorithms (trees, graphs, hash tables, NP-complete problems, Huffman coding)
 - Systems Programming (file systems, processes, memory management, shell scripting)
 - Object-Oriented Design (class hierarchies, interfaces, inheritance, exceptions, generics)
 - Computer Vision (robotics vision pipelines, autonomous navigation)
 - Signal Processing (audio effects, quantization, noise, synthesis)
 - Full-Stack Web Development (frontend design, backend APIs, database integration, authentication)
 - Embedded & Hardware Systems (soldering, wiring, pneumatics, motor control, high-voltage electronics)
-

Professional Experience

Roxy 14 Movie Theatre — *Concessions Attendant / Usher / Doorman*
2022 – Present

- Served hundreds of customers per night while upholding service and cleanliness standards
- Managed opening/closing duties ensuring daily operational stability
- Prepared food safely and efficiently, maintaining compliance with food safety regulations

- Provided front-facing customer service as theatre greeter and information guide
-

Project & Major Experience

MCHS Robotics Team (FRC) — *Lead Programmer & Treasurer* 2022 – 2024

- Managed team budget of \$8,000 and organized club events
- Designed and implemented robot systems: wiring, pneumatics, motors, and sensors
- Programmed drive systems (tank, swerve, autonomous with gyroscope), vision systems, and robotic arms in **Java**
- Mentored new members with technical presentations on robotics programming and engineering

Coding & Game Development Clubs (MCHS) 2021 – 2024

- Organized and taught beginner coding/game dev workshops
- Participated in hackathons and group software projects

Everybody Just Wants One More Day (Class Project) — *Unity Developer*

- Collaborated with a cross-disciplinary team of 10+ artists, writers, and programmers
- Developed 2D movement systems and event triggers in Unity (C#)
- Coordinated hosting and deployment for a game viewed by 200+ people

Imersi (Personal Project) — *Full-Stack Developer*

- Built a **MERN stack** web-based quiz game with live updates via WebSockets
- Implemented secure user accounts with MongoDB and session management (cookies & Redis)

Audio Programming Projects (C++)

- **fQuantizer**: VST3 plugin that applies scalable quantization to amplitude values, simulating reduced bit-depth audio
- **Noise**: VST3 plugin adding scalable noise effects for sound design

Hardware & Embedded Projects

- **Tesla Coil (15kV)**: Designed coils, capacitor bank; maintained safe procedures for high-voltage systems
 - **Medical Display Robot**: Raspberry Pi rover streaming vitals & heart-rate monitor via remote-access website
 - **Jukebot**: Raspberry Pi media system; Discord bot retrieves and queues videos with yt-dlp for group playback
-

Activities & Memberships

- **Cal Poly Robotics Club** — Active member, contributing to robotics projects and competitions
- **Cal Poly Audio Engineering Society** — Participate in building speakers, amplifiers, synthesizers, FX pedals, and sound systems
- **MCHS Robotics Team** — Former lead programmer and treasurer (FRC competition)
- **MCHS Coding & Game Development Clubs** — Workshop leader and hackathon participant